Experimentation & Evaluation

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Lecture 04

Measurements

Three aims of research

Valid
Actually shows what you intended to demonstrate

Reliable
Can be repeated (by yourself) and reproduced (by others)
Can be generalized beyond particular experiment

Important
Subjective, but good to strive for lasting contribution
Invalid/unreliable is always unimportant

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Maximizing reliability

Produce precise measures
Minimize experimental imperfections, produce the same results under the same conditions

Have an objective definition of what is measured



What is being measured?

Researcher A found that when giving 20 subjects the new drug NOGOOD, they all had a heart attack.

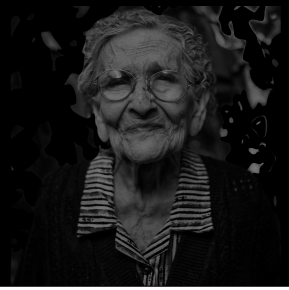
Researcher B found that when giving 20 subjects the new drug NOGOOD, no one had heart attack.

What is being measured?

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Researcher B found that when giving 20 subjects the new drug NOGOOD, no one had heart attack.

Impossible to replicate

Researcher A found that when giving 20 subjects the new drug NOGOOD, they all had a heart attack.

Researcher B found that when giving 20 subjects the new drug NOGOOD, no one had heart attack.

Maximizing reliability

Produce precise measures
Minimize experimental imperfections, produce the same results under the same conditions

Have an objective definition of what is measured
Often clear-cut
- E.g., execution time as a measure for performance

Definition by consensus
- Experts agree on “true instances”

Provide operational definition
- E.g., “in this study, I define code complexity as the number of decision points implemented”



Non-parametric scales of measurements

Nominal (categorical) scale
Set of categories (names) with no order, e.g., male and female, colors
- Very little statistics can be performed, it’s mostly about counting

Ordinal scale
Set of ordered categories (sequence)
- Ranked observation, e.g., small, medium, large
- Slightly better in terms of statistics, we have a direction for differences but not a magnitude

Parametric scales of measurements

Interval scale
Ordered categories in intervals of the same size
- Difference in number equals difference in magnitude, e.g., temperature
- Arbitrary zero point, does not mean absence of value

Ratio scale
Interval scale with real zero point
- Time, distance, weight, #errors
- Allows for meaningful rations (e.g., twice as fast, half as far)

Assignment 01

Code Performance

The Bubble Algorithm Library

Who asked?

Bubble Inc., implementer of a class library implementing a sorting algorithm

Purpose?

Decide which of four sort algorithm implementations to include in their library

Questions?

Which implementations run faster (for which kind of inputs)?

Audience?

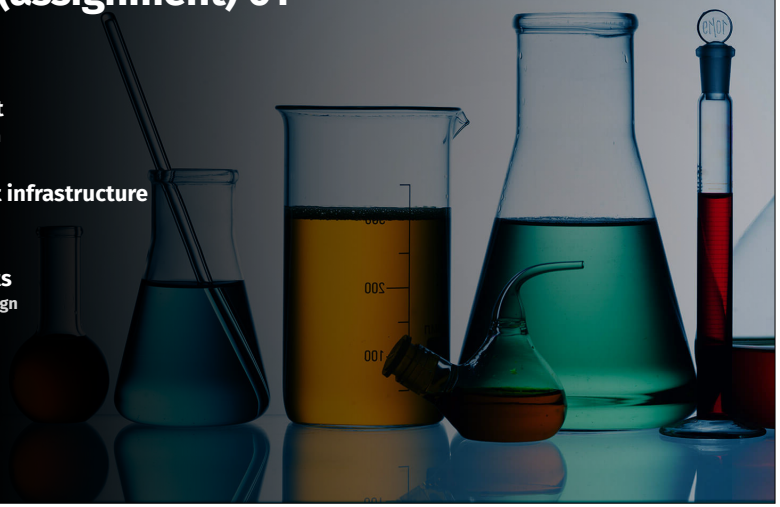
Java developers

Sort algorithm implementation

A class implementing the **Sorter** interface

```
interface Sorter<T extends Comparable<T>> {
  void sort(T[] items);
}
```

Experiment (assignment) 01



Design the experiment
Try to be as precise as you can

Develop measurement infrastructure
Clean code, documented

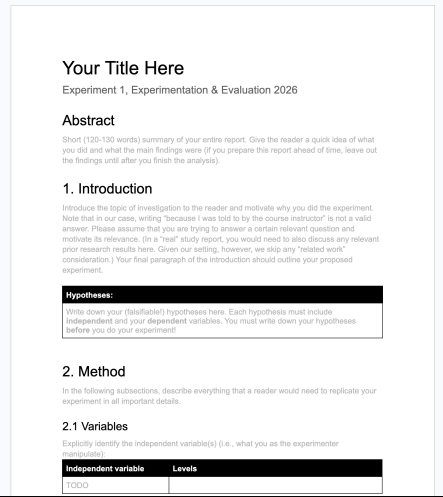
Perform measurements
Follow your experimental design

Create report
Follow template in iCorsi

Experiment Report

Copy the Google Doc

https://docs.google.com/document/d/1HO0JJgxPk6yLzUBy0lvHTZ-2GWu7FHLUNJz-8wTztos/edit?usp=sharing



Your Title Here

Experiment 1, Experimentation & Evaluation 2026

Abstract

Short (120-130 words) summary of your entire report. Give the reader a quick idea of what you did and what the main findings were (if you prepare this report ahead of time, leave out the findings until after you finish the analysis).

1. Introduction

Introduce the topic of investigation to the reader and motivate why you did the experiment. Note that in our case, writing "because I was told to by the course instructor" is not a valid answer. Please assume that you are trying to answer a certain relevant question and motivate its relevance. (In a "real" study report, you would need to also discuss any relevant prior research results here. Given our setting, however, we skip any "related work" consideration.) Your final paragraph of the introduction should outline your proposed experiment.

Hypotheses:
Write down your (falsifiable!) hypotheses here. Each hypothesis must include **independent** and your **dependent** variables. You must write down your hypotheses **before** you do your experiment!

2. Method

In the following subsections, describe everything that a reader would need to replicate your experiment in all important details.

2.1 Variables

Explicitly identify the independent variable(s) (i.e., what you as the experimenter manipulate):

Independent variable	Levels
TODO	

Important considerations

Dependent variables

Performance. But what exactly? How to measure? Their scale.

Independent variables

The algorithm you pick with its four possible levels (the type of algorithm).
This is know as the "factor" of the experiment.
You can have more than just the algorithm, you may want to play with the size of the array to sort?

Confounding factors

Control variables (keep constant), e.g., the load on your computer.